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23R-D25 (MAT5204C)

2025

MATHEMATICS

(MAJOR – II)

Paper : MAT5204C

(Multivariable Calculus)

Full Marks : 60

Time : 2½ hours

The figures in the margin indicate full marks for the questions.

1. Write the answer of the following questions :

1×7=7

(a) If $f(x, y) = x^2y$, then what is the derivative of f at $(1, 1)$?

(b) If $f(x, y) = x^3y \sin^{-1}(5y/x)$, then what is the value of n such that

$$x \frac{\partial f}{\partial x} + y \frac{\partial f}{\partial y} = nf ?$$

E16FO 0047 $x^3 x^2 y \sin^{-1}(5y/x)$
 $+ y^2$

Contd.

- (c) Is $(0,0)$ a saddle point of $f(x,y) = x^2 - y^2$? Justify.
- (d) If $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$, then what are the values $\text{div}(\vec{r})$ and $\text{curl}(\vec{r})$?
- (e) If C is the circle $x^2 + y^2 = 1$ oriented positively, then what is the value of $\oint_C (3y - e^{\sin x})dx + (3x + \sqrt{y^2 + 1})dy$ due to Greene's Theorem?
- (f) If S is the surface of the unit sphere $x^2 + y^2 + z^2 = 1$ and $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$, then what is the value of $\iint_S \vec{r} \cdot d\vec{S}$ due to Gauss Divergence Theorem?
- (g) Is the statement "If $\vec{F}$ is a conservative field, then $\vec{F}$ is irrotational." true? Justify.

2. Give short answer to the following questions : 2×4=8

- (a) Describe all level surfaces of the function $f: \mathbb{R}^3 \rightarrow \mathbb{R}$ given by $f(x,y,z) = x^2 + y^2$.

✓(b) If $z = xy^2$, $x = \cos t$, and $y = \sin t$, then find $\frac{dz}{dt}$ at $t = \frac{\pi}{2}$.

(c) Changing the order of integration, find the value of $\int_0^\pi \int_x^\pi \frac{\sin y}{y} dy dx$.

(d) Find a relation among a, b , and c such that $\vec{F} = (ax + y)\hat{i} + (by - 3z)\hat{j} + (3x - cz)\hat{k}$ is a solenoidal.

3. Answer (a) and [**either** (b) **or** (c)] and [**either** (d) **or** (e)] from the following: $5 \times 3 = 15$

(a) If $x = r \cos \theta$ and $y = r \sin \theta$, then show

$$\text{that } \frac{\partial^2 \theta}{\partial x \partial y} = -\frac{\cos 2\theta}{r^2}.$$

✓(b) Find the equation of the tangent plane and the normal line to the graph of the

function $z = x^2 y + \frac{1}{1+y^2}$ at the point

$\left(1, 1, \frac{3}{2}\right)$. If the tangent plane passes through the point $(2, 2, t)$, then find values of t .

- (c) Substituting $u = x - y$ and $v = x + y$, find the value of

$$\iint_D (x - y)^2 \sin^2(x + y) dx dy,$$

where D is the parallelogram with vertices $(\pi, 0), (2\pi, \pi), (\pi, 2\pi), (0, \pi)$.

- (d) Find the surface area of the cone $z = \sqrt{x^2 + y^2}$ with $0 \leq z \leq 1$.

- (e) Find the volume of the region D enclosed by the surfaces $z = x^2 + 3y^2$ and $z = 8 - x^2 - y^2$.

4. Answer **either** (a) **or** (b) of the following:

10

- ✓ (a) Consider the function $f: \mathbb{R}^2 \rightarrow \mathbb{R}$ given

$$\text{by } f(x, y) = \begin{cases} \frac{x^3}{x^2 + y^2}, & \text{if } (x, y) \neq (0, 0); \\ 0, & \text{if } (x, y) = (0, 0). \end{cases}$$

Examine

- (i) the continuity of f at the origin.
- (ii) the existence of all partial derivatives of f at the origin.
- (iii) the differentiability of f at the origin.

(b) Let $f: \mathbb{R}^2 \rightarrow \mathbb{R}$ be a function and $(a, b) \in \mathbb{R}^2$ such that $f_{xy}(a, b)$ and $f_{yz}(a, b)$ exist in a neighborhood of the point (a, b) . If the mixed partial derivatives f_{xy} and f_{yz} are continuous at (a, b) , then show that

$$f_{xy}(a, b) = f_{yz}(a, b).$$

If
$$f(x, y) = \begin{cases} \frac{xy}{x+y}, & \text{if } (x, y) \neq (0, 0); \\ 0, & \text{if } (x, y) = (0, 0); \end{cases}$$

then examine if $f_{xy}(0, 0) = f_{yz}(0, 0)$.

5. Answer **either** (a) **or** (b) of the following:

10

✓(a) Define a critical point of a function $f: \mathbb{R}^2 \rightarrow \mathbb{R}$. Find all critical points of the function

$f(x, y) = x^3 + y^3 - 3x - 12y + 20$, and then classify them. Find extreme values of the function, if exist.

- (b) Using Lagrange's multiplier method, find extreme values of the function $f(x, y) = 3x + 4y$ on the circle $x^2 + y^2 = 1$ occur at the points (α, β) , then find the values of $2(\alpha^2 + \beta^2)$.

6. Answer **either** (a) **or** (b) of the following: 10

- (a) Show that

$\vec{F} = (e^x \cos y + yz)\hat{i} + (xz - e^x \sin y)\hat{j} + (xy + z)\hat{k}$ is conservative. Find the corresponding potential function.

Using this, find the work done by $\vec{F}$

in moving from $(0, 0, 2)$ to $(0, \frac{\pi}{2}, 0)$.

- (b) State the Stokes' Theorem. Verify the Stokes' Theorem for the function

$\vec{F} = y\hat{i} - x\hat{j}$ over the hemisphere

$S: x^2 + y^2 + z^2 = 9, z \geq 0$, with its

bounding circle $C: x^2 + y^2 = 9, z = 0$.